

2026 ICU & EMERGENCY CLINICAL POCKET GUIDE

Evidence-Based Decision Rules, Ventilation Targets, Drug Infusions & Emergency Formulas

1. SEPSIS-3 & HEMODYNAMIC RESUSCITATION PROTOCOL

- **qSOFA Score (Bedside Triage):** 1 pt each for RR ≥ 22 /min, Altered Mental Status (GCS <15), SBP ≤ 100 mmHg.
Interpretation: Score ≥ 2 indicates high risk of prolonged ICU stay & mortality -> Initiate Sepsis Bundle immediately.
- **SOFA Score (ICU Organ Dysfunction):** Evaluates 6 systems (PaO₂/FiO₂, Platelets, Bilirubin, MAP/Vasopressors, GCS, Cr/Urine).
An acute increase in SOFA ≥ 2 points indicates organ dysfunction secondary to infection.
- **Mean Arterial Pressure (MAP) Equation:** $MAP = (2 * DBP + SBP) / 3$
Resuscitation Target: MAP ≥ 65 mmHg in Septic Shock.
First-line Vasopressor: Norepinephrine (0.02 - 0.5 mcg/kg/min). Add Vasopressin at 0.03 units/min if refractory.
- **Initial Fluid Resuscitation:** Administer 30 mL/kg balanced crystalloids (Plasmalyte or LR) within 3 hours for lactate ≥ 4 mmol/L.

2. ARDS BERLIN DEFINITION & MECHANICAL VENTILATION

- **PaO₂/FiO₂ (P/F Ratio) Classification (PEEP ≥ 5 cmH₂O):**

 - Mild ARDS: $200 < P/F \leq 300$ mmHg | Mortality ~27%
 - Moderate ARDS: $100 < P/F \leq 200$ mmHg | Mortality ~32%
 - Severe ARDS: $P/F \leq 100$ mmHg | Mortality ~45% (Consider Prone Positioning & NMBA)

- **Protective Lung Ventilation Target:**
 - Initial Tidal Volume: 6 mL/kg Predicted Body Weight (PBW).
 - PBW Male = $50 + 0.91 * (\text{Height cm} - 152.4)$ | PBW Female = $45.5 + 0.91 * (\text{Height cm} - 152.4)$
 - Plateau Pressure (Pplat) Target: ≤ 30 cmH₂O.
 - Driving Pressure (Delta P = Pplat - PEEP) Target: $< 13 - 15$ cmH₂O.

3. NEUROLOGICAL ASSESSMENT & TRAUMA (GCS)

- **Glasgow Coma Scale (GCS) Total: 3 to 15 Points**
 - Eye Response (E 1-4): Spontaneous (4), To Speech (3), To Pain (2), None (1).
 - Verbal Response (V 1-5): Oriented (5), Confused (4), Inappropriate (3), Incomprehensible (2), None (1).
 - Motor Response (M 1-6): Obeys Commands (6), Localizes Pain (5), Withdraws (4), Flexion (3), Extension (2), None (1).
- **Clinical Rule:** GCS ≤ 8 -> Intubate for Airway Protection.

4. RENAL FUNCTION & ACID-BASE DIAGNOSTICS

- Cockcroft-Gault CrCl = $[(140 - \text{Age}) * \text{Weight kg}] / [72 * \text{Serum Cr mg/dL}]$ (* 0.85 for Females)
- CKD-EPI 2021 GFR: Race-free equation recommended by KDIGO 2026 guidelines.
- Serum Anion Gap = $\text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-)$ | Normal Range: 8 - 12 mEq/L
 - High Anion Gap Metabolic Acidosis (HAGMA): MUDPILES / GOLDMARK (Lactate, DKA, Uremia, Toxins).
- Winter's Formula (Expected PCO₂ in Metabolic Acidosis): $\text{PCO}_2 = 1.5 * [\text{HCO}_3^-] + 8 \pm 2$
- Sodium Correction in Hyponatremia: Max correction rate $\leq 8 - 10$ mEq/L in 24 hours to prevent ODS.

5. CARDIOLOGY & ANTICOAGULATION SCORE REFERENCE

- CHA₂DS₂-VASc Stroke Risk in AFib: CHF (1), HTN (1), Age ≥ 75 (2), Diabetes (1), Stroke/TIA (2), Vascular (1), Age 65-74 (1), Sex (1)
Interpretation: Score ≥ 2 in Males or ≥ 3 in Females -> Anticoagulation recommended (DOAC 1st line).
- HAS-BLED Bleeding Risk: HTN (1), Abnormal Renal/Liver (1 or 2), Stroke (1), Bleeding (1), Labile INR (1), Elderly (1), Drugs/Alcohol (1)
Score ≥ 3 indicates high bleeding risk; warrants close monitoring rather than withholding anticoagulation.

6. CRITICAL CARE DRUG INFUSION & STEROID CONVERSION TABLE

- Vasopressor Starting Doses:
 - Norepinephrine: 0.02 - 0.05 mcg/kg/min (Titrate q2-5min for MAP ≥ 65)
 - Vasopressin: Fixed 0.03 units/min (Second-line non-titratable adjunct)
 - Epinephrine: 0.01 - 0.5 mcg/kg/min (Inotrope + Vasopressor)
- Systemic Steroid Equivalent Doses:
 - Hydrocortisone: 20 mg (Short acting, high mineralocorticoid activity)
 - Prednisone / Prednisolone: 5 mg
 - Methylprednisolone: 4 mg
 - Dexamethasone: 0.75 mg (Long acting, zero mineralocorticoid activity)